

DSA Lab Sheet

II Year / II Part

Faculty: Computer and Electronics

Labsheet #1: Revision of C Programming

Objectives:

1. Understand and use different types of functions in C
2. Work with arrays and perform operations on them
3. Use pointers effectively to access and manipulate data
4. Understand the relationship between arrays and pointers

Theory:

An array is a collection of elements of the same data type stored in contiguous memory locations.

Functions divide a program into smaller, modular parts. Types:

1. Without arguments & no return value
2. With arguments & no return value
3. Without arguments & with return value
4. With arguments & with return value

A pointer is a variable that stores the address of another variable.

Tasks:

1 WAP in C using structure that takes input Name, DOB, RollNo and marks of 5 subjects from user and display the information to the user with additional informations: age, average marks and percentage.

Requirements: Separate functions for each of following tasks

1. greet() function that greets user in console as soon as it is called.
2. getInfo() function that take information from user and return the obtained info to the main function.
3. calculateAge() function that takes DOB as input and returns back age
4. calculateAverage() function that takes array of marks and return back average marks.
5. calculatePercent() function that takes in average and return back percentage.
6. displayResult() function that takes structure as input and display all necessary result to user.

2. WAP to reverse an array using function.
3. WAP with a function to find the maximum and minimum from an array using pointers.

```
void findMaxMin(int *arr, int size, int *max, int *min) {  
    ...  
}
```

4. WAP in C to Swap two integers using:
 1. Call by value
 2. Call by reference
5. WAP in C to find factorial of an integer using:
 1. Iteration
 2. Recursion